NAME SURNAME: STUDENT ID:

	A Gazi University Industrial Engineering Department 2025-2026 Spring Semester **Database Management Systems (DBMS)** Final Exam A	Signature /İmza

Rules: Please read questions **carefully**. Write **clearly** in the spaces provided. SQL answers must be **valid** PostgreSQL-style SQL. Equivalent SQL **might** be accepted if it returns exactly the requested result. **Do not** use notes, books, phones, or electronic devices. **Turn off** your mobile phones and leave it facing down on your desk. Put your student ID **on the desk** during the exam. Leaving without submitting the question paper is prohibited.

Good Luck **Dr Ercan Ezin**

QUESTION 1

A. Define or explain following DBMS terms objectively: **primary key**, **foreign key**, and **unary relationship**. Write each definition in one or two precise sentences. (3+3+4=**10 Points total**)

B. Design an ER diagram below for a car mechanic shop using the standard notation. The shop is run by *Mechanics*, who have a unique **EmployeeID**, a **Specialty**, and a **Full Name** (composed of First and Last names). Each mechanic can **mentor** multiple other mechanics (mentor-mentee relationship), though a junior mechanic is assigned to at most one mentor. Mechanics perform **Repairs** on **Vehicles**; because a vehicle may be serviced by multiple mechanics over time and a mechanic services many vehicles, you must record the **VisitDate** and **Cost** for every specific repair visit day. *Vehicles* are identified by a unique **VIN** (Vehicle Identification Number) and a **Year**. Finally, the shop tracks the *Owners* for each vehicle; in this system, an owner is treated as a **weak entity** that cannot exist in the database without their registered primary vehicle. For each owner, record their **TCID** (as partial identifier), **FirstName,** and a Multi-Value **PhoneNumber**. Your diagram must clearly show all entities (**3 points**), attributes (identifying primary and partial keys) (**4 points**), relationships (including recursive and weak) (**4 points**), and **cardinality ratios** (1:1, 1:N, or M:N)(**4 points**) correctly and most effectively to get full mark. (**15 P**)

NAME SURNAME: STUDENT ID:

	B Gazi University Industrial Engineering Department 2025-2026 Spring Semester **Database Management Systems (DBMS)** Final Exam B	Signature /İmza

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Good Luck **Dr Ercan Ezin**

QUESTION 1

A. Define or explain following DBMS terms objectively: **table attribute**, **varchar type**, and **weak relationship**. Write each definition in one or two precise sentences. (3+3+4=**10 Points total**)

B. Design an ER diagram below for a car mechanic shop using the standard notation. The shop is run by *Mechanics*, who have a unique **EmployeeID**, a **Specialty**, and a **Full Name** (composed of First and Last names). Each mechanic can **mentor** multiple other mechanics (mentor-mentee relationship), though a junior mechanic is assigned to at most one mentor. Mechanics perform **Repairs** on **Vehicles**; because a vehicle may be serviced by multiple mechanics over time and a mechanic services many vehicles, you must record the **VisitDate** and **Cost** for every specific repair visit day. *Vehicles* are identified by a unique **VIN** (Vehicle Identification Number) and a **Year**. Finally, the shop tracks the *Owners* for each vehicle; in this system, an owner is treated as a **weak entity** that cannot exist in the database without their registered primary vehicle. For each owner, record their **FirstName** (as partial identifier)**, TCID** and a Multi-Value **PhoneNumber**. Your diagram must clearly show all entities (**3 points**), attributes (identifying primary and partial keys) (**4 points**), relationships (including recursive and weak) (**4 points**), and **cardinality ratios** (1:1, 1:N, or M:N)(**4 points**) correctly and most effectively to get full mark. (**15 P**)

QUESTION 2: Note that this question is not related to the Question 1. Use this single table to answer **A, B, C, D**.

job_id	line_code	machine_type	priority	opened_on	closed_on	cost	status	technician_note
1	L1	CNC Router	HIGH	2026-04-15T10:00:00.000Z	(empty)	1500.00	OPEN	(empty)
2	L2	Hydraulic Press	MEDIUM	2026-04-20T11:00:00.000Z	(empty)	5000.00	WAITING	(empty)
3	L1	Conveyor	LOW	2026-05-10T09:00:00.000Z	(empty)	1200.00	OPEN	Needs parts
4	L3	CNC Router	HIGH	2026-05-12T14:00:00.000Z	(empty)	9000.00	OPEN	(empty)
5	L3	Drill Press	MEDIUM	2026-05-15T08:00:00.000Z	(empty)	400.00	OPEN	Routine check
6	L5	Drill Press	LOW	2026-05-16T09:00:00.000Z	(empty)	300.00	OPEN	Excluded due to LOW
7	L1	Conveyor	MEDIUM	2026-03-01T08:00:00.000Z	2026-03-02T12:00:00.000Z	6000.00	CLOSED	Fixed motor
8	L1	CNC Router	HIGH	2026-03-05T09:00:00.000Z	2026-03-06T17:00:00.000Z	5500.00	CLOSED	Replaced bearing
9	L2	Hydraulic Press	HIGH	2026-03-10T10:00:00.000Z	2026-03-11T11:00:00.000Z	4000.00	DONE	Alternative status na
10	L4	Injection Molder	CRITICAL	2026-04-28T07:00:00.000Z	(empty)	2500.00	OPEN	Pending review

A. Write a table creation SQL command to generate an empty table named *maintenance_job* with attributes and data types given above. Use your judgment to assign the most suitable max or min value for table attributes where it applies. (**5P**)

B. Write an SQL command for *maintenance_job* table to return **job_id, line_code**, and **cost** for jobs whose status is **OPEN** or **WAITING**, whose cost is between **1000.00** and **8000.00**, and whose **technician_note** is either **NULL** or an **empty string**. Sort by highest **cost**, then smallest **job_id**. (**10 P**)

C. Write an SQL command to return distinct **machine_type** values for line codes **L1**, **L3**, or **L5**, excluding jobs with **priority LOW**. Sort result alphabetically. (**10 P**)

D. For each **line_code** write an SQL command to show the number of closed jobs and the total closed job cost. Include only jobs where **closed_on** is not **NULL or empty**. Show only lines whose total closed job cost is greater than 10000. Sort by total cost from high to low. **(10P)**

QUESTION 3

Use the following tables and rows to answer **A, B, C, D.**

ws_id	station_name	area
10	Cutting	North
20	Assembly	East
30	Packaging	East

workstation

order_id	ws_id	product_code	planned_qty	due_date
101	10	AX100	500	2026-06-05T00:00:00.000Z
102	20	BX200	300	2026-06-08T00:00:00.000Z
103	20	AX100	200	2026-06-12T00:00:00.000Z
104	30	CX300	600	2026-06-09T00:00:00.000Z
105	(empty)	DX400	150	2026-06-15T00:00:00.000Z

production_order

check_id	order_id	checked_qty	defects	inspector
1	101	500	5	Ada
2	102	280	14	Bora
3	103	200	0	Ada
4	104	600	30	Deniz

quality_check

A. Write an SQL command to list **order_id**, **product_code**, **due_date**, and **station_name** for orders due before 2026-06-10. Include only orders assigned to a workstation. Sort by **due_date**. **(5P)**

B. Fill the output table on the right with values that are produced by this SQL command. **(5P)**

```
SELECT w.area, COUNT(o.order_id) AS order_count
FROM workstation w
LEFT JOIN production_order o ON w.ws_id = o.ws_id
GROUP BY w.area
ORDER BY w.area;
```

area	order_count

C. Write an SQL command to show **product_code**, total planned quantity, and total defects for checked orders only. Group by **product** and show only products with more than 10 total defects. Sort by total defects descending. (**10P**)

D. Write the SQL command and fill the exact output table below. For each workstation area, compute

$$defect\_rate = total\ defects\ /total\ checked\ quantity$$

Include only areas with at least **500** checked units. Round the rate to 4 decimal places and sort by highest rate. Use PostgreSQL ROUND(expression, 4) to round the defect_rate. (**20P**)

OUTPUT			
area	total_checked	total_defects	defect_rate